

12.1 How to read this chapter

The USB host controller is available on the LPC1768, LPC1766, LPC1765, LPC1758, LPC1756, and LPC1754. On these devices, the USB controller can be configured for device, Host, or OTG operation.

12.2 Basic configuration

The USB controller is configured using the following registers:

1. Power: In the PCONP register ([Table 46](#)), set bit PCUSB.
Remark: On reset, the USB block is disabled (PCUSB = 0).
2. Clock: The USB block can be used with a dedicated USB PLL (PLL1) to obtain the USB clock or with the Main PLL (PLL0). See [Section 4.6.1](#).
3. Pins: Select USB pins and their modes in PINSEL0 to PINSEL5 and PINMODE0 to PINMODE5 ([Section 8.5](#)).
4. Wake-up: Activity on the USB bus port can wake up the microcontroller from Power-down mode, see [Section 4.8.8](#).
5. Interrupts: Interrupts are enabled in the NVIC using the appropriate Interrupt Set Enable register.
6. Initialization: see [Section 13.11](#).

12.3 Introduction

This section describes the host portion of the USB 2.0 OTG dual role core which integrates the host controller (OHCI compliant), device controller, and I²C interface. The I²C interface controls the external OTG ATX.

The USB is a 4 wire bus that supports communication between a host and a number (127 max.) of peripherals. The host controller allocates the USB bandwidth to attached devices through a token based protocol. The bus supports hot plugging, un-plugging and dynamic configuration of the devices. All transactions are initiated by the host controller.

The host controller enables data exchange with various USB devices attached to the bus. It consists of register interface, serial interface engine and DMA controller. The register interface complies to the OHCI specification.

Table 253. USB (OHCI) related acronyms and abbreviations used in this chapter

Acronym/abbreviation	Description
AHB	Advanced High-Performance Bus
ATX	Analog Transceiver
DMA	Direct Memory Access
FS	Full Speed

Table 253. USB (OHCI) related acronyms and abbreviations used in this chapter ...continued

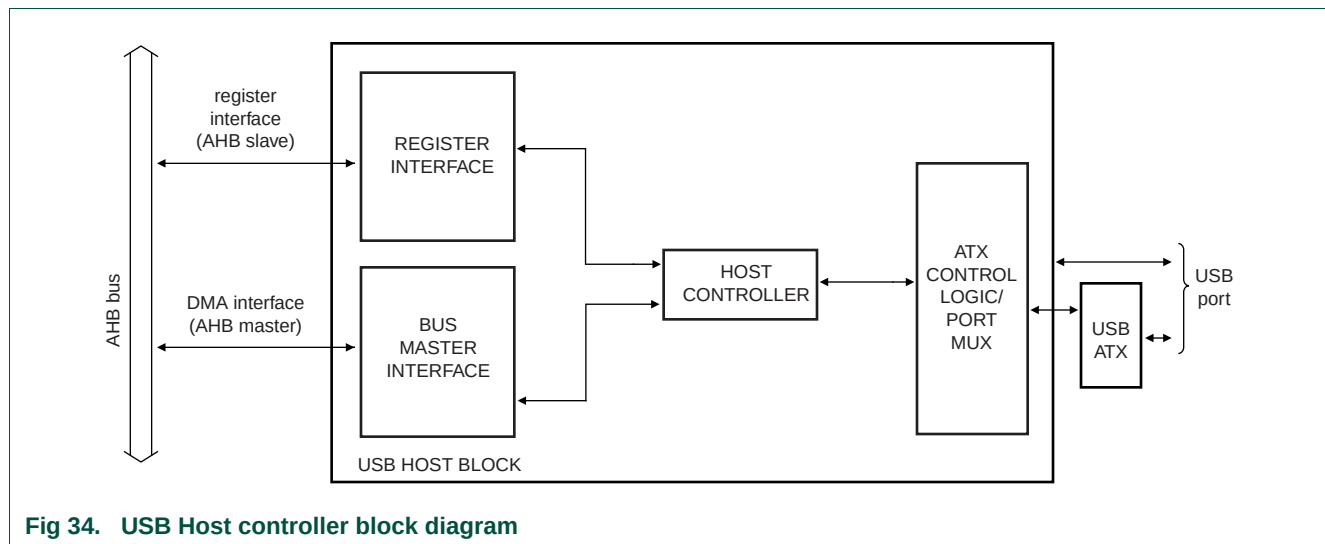
Acronym/abbreviation	Description
LS	Low Speed
OHCI	Open Host Controller Interface
USB	Universal Serial Bus

12.3.1 Features

- OHCI compliant.
- OpenHCI specifies the operation and interface of the USB Host Controller and SW Driver
 - USBOperational: Process Lists and generate SOF Tokens.
 - USBReset: Forces reset signaling on the bus, SOF disabled.
 - USBSuspend: Monitor USB for wake-up activity.
 - USBResume: Forces resume signaling on the bus.
- The Host Controller has four USB states visible to the SW Driver.
- HCCA register points to Interrupt and Isochronous Descriptors List.
- ControlHeadED and BulkHeadED registers point to Control and Bulk Descriptors List.

12.3.2 Architecture

The architecture of the USB host controller is shown below in [Figure 34](#).



12.4 Interfaces

The USB interface is controlled by the OTG controller. It has one USB port.

12.4.1 Pin description

Table 254. USB Host port pins

Pin name	Direction	Description	Type
USB_D+	I/O	Positive differential data	USB Connector
USB_D-	I/O	Negative differential data	USB Connector
USB_UP_LED	O	GoodLink LED control signal	Control
USB_PPWR	O	Port power enable	Host power switch
USB_PWRD	I	Port power status	Host power switch
USB_OVRCR	I	Over-current status	Host power switch

12.4.1.1 USB host usage note

The USB block can be configured as USB host. For details on how to connect the USB port, see the USB OTG chapter, [Section 13.7](#).

The USB device/host/OTG controller is disabled after RESET and must be enabled by writing a 1 to the PCUSB bit in the PCONP register, see [Table 46](#).

12.4.2 Software interface

The software interface of the USB host block consists of a register view and the format definitions for the endpoint descriptors. For details on these two aspects see the OHCI specification. The register map is shown in the next subsection.

12.4.2.1 Register map

The following registers are located in the AHB clock 'cclk' domain. They can be accessed directly by the processor. All registers are 32 bits wide and aligned in the word address boundaries.

Table 255. USB Host register address definitions

Name	Address	R/W ⁽¹⁾	Function	Reset value
HcRevision	0x5000 C000	R	BCD representation of the version of the HCI specification that is implemented by the Host Controller.	0x10
HcControl	0x5000 C004	R/W	Defines the operating modes of the HC.	0x0
HcCommandStatus	0x5000 C008	R/W	This register is used to receive the commands from the Host Controller Driver (HCD). It also indicates the status of the HC.	0x0
HcInterruptStatus	0x5000 C00C	R/W	Indicates the status on various events that cause hardware interrupts by setting the appropriate bits.	0x0
HcInterruptEnable	0x5000 C010	R/W	Controls the bits in the HcInterruptStatus register and indicates which events will generate a hardware interrupt.	0x0
HcInterruptDisable	0x5000 C014	R/W	The bits in this register are used to disable corresponding bits in the HcInterruptStatus register and in turn disable that event leading to hardware interrupt.	0x0
HcHCCA	0x5000 C018	R/W	Contains the physical address of the host controller communication area.	0x0
HcPeriodCurrentED	0x5000 C01C	R	Contains the physical address of the current isochronous or interrupt endpoint descriptor.	0x0

Table 255. USB Host register address definitions ...continued

Name	Address	R/W ^[1]	Function	Reset value
HcControlHeadED	0x5000 C020	R/W	Contains the physical address of the first endpoint descriptor of the control list.	0x0
HcControlCurrentED	0x5000 C024	R/W	Contains the physical address of the current endpoint descriptor of the control list	0x0
HcBulkHeadED	0x5000 C028	R/W	Contains the physical address of the first endpoint descriptor of the bulk list.	0x0
HcBulkCurrentED	0x5000 C02C	R/W	Contains the physical address of the current endpoint descriptor of the bulk list.	0x0
HcDoneHead	0x5000 C030	R	Contains the physical address of the last transfer descriptor added to the 'Done' queue.	0x0
HcFmInterval	0x5000 C034	R/W	Defines the bit time interval in a frame and the full speed maximum packet size which would not cause an overrun.	0x2EDF
HcFmRemaining	0x5000 C038	R	A 14-bit counter showing the bit time remaining in the current frame.	0x0
HcFmNumber	0x5000 C03C	R	Contains a 16-bit counter and provides the timing reference among events happening in the HC and the HCD.	0x0
HcPeriodicStart	0x5000 C040	R/W	Contains a programmable 14-bit value which determines the earliest time HC should start processing a periodic list.	0x0
HcLSThreshold	0x5000 C044	R/W	Contains 11-bit value which is used by the HC to determine whether to commit to transfer a maximum of 8-byte LS packet before EOF.	0x628h
HcRhDescriptorA	0x5000 C048	R/W	First of the two registers which describes the characteristics of the root hub.	0xFF000902
HcRhDescriptorB	0x5000 C04C	R/W	Second of the two registers which describes the characteristics of the Root Hub.	0x60000h
HcRhStatus	0x5000 C050	R/W	This register is divided into two parts. The lower D-word represents the hub status field and the upper word represents the hub status change field.	0x0
HcRhPortStatus[1]	0x5000 C054	R/W	Controls and reports the port events on a per-port basis.	0x0
HcRhPortStatus[2]	0x5000 C058	R/W	Controls and reports the port events on a per port basis.	0x0
Module_ID/Ver_Rev_ID	0x5000 C0FC	R	IP number, where yy (0x00) is unique version number and zz (0x00) is a unique revision number.	0x3505yyzz

[1] The R/W column in [Table 255](#) lists the accessibility of the register:

- Registers marked 'R' for access will return their current value when read.
- Registers marked 'R/W' allow both read and write.

12.4.2.2 USB Host Register Definitions

Refer to the OHCI specification document on the Compaq website for register definitions.